

# Wednesday Warm Up: Current, Ohm's Law, and Resistance

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## 1 Memory Bank

- $I = \Delta Q / \Delta t$  ... Definition of current.
- $\Delta V = IR$  ... Ohm's Law.
- $R = \rho(L/A)$  ... Resistivity,  $\rho$ , and resistance,  $R$ , for a conductor with cross-sectional area  $A$  and length  $L$ .
- $\rho = \rho_0(1 + \alpha\Delta T)$  ... Resistivity depends on changes in temperature.
- $R = R_0(1 + \alpha\Delta T)$  ... Resistance depends on changes in temperature.

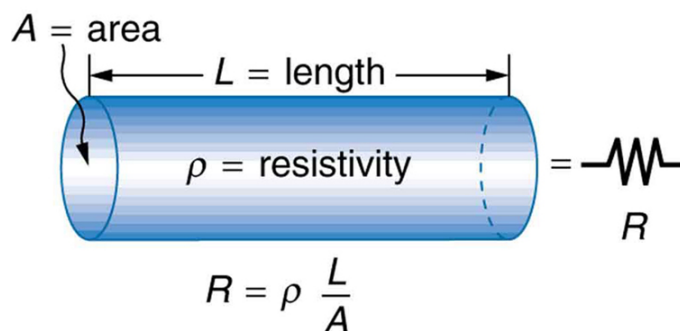


Figure 1: Resistivity and resistance in a conductor.

## 2 Current

1. The 200-A current through a spark plug moves 0.300 mC of charge. How long does the spark last?

2. Electron guns are used in x-ray tubes. The electrons are accelerated through a relatively large voltage and directed onto a metal target, producing X-rays. (a) How many electrons per second strike the target if the current is 0.500 mA? (b) What charge strikes the target in 0.750 s?

2. How many volts are supplied to operate an indicator light on a DVD player that has a resistance of  $140\Omega$ , given that 25.0 mA passes through it?

## 4 Resistance and Resistivity

1. If the 0.010 mm diameter tungsten filament in a light bulb is to have a resistance of  $0.200\Omega$  at  $20.00^\circ\text{C}$ , how long should it be? The resistivity of tungsten is  $5.6 \times 10^{-8}\Omega\text{ m}$ .

## 3 Ohm's Law

1. (a) Find the voltage drop in an extension cord having a  $0.0600\Omega$  resistance and through which 5.00 A is flowing. (b) A cheaper cord utilizes thinner wire and has a resistance of  $0.300\Omega$ . What is the voltage drop in it when 5.00 A flows? (c) Why is the voltage to whatever appliance is being used reduced by this amount? What is the effect on the appliance?

2. The temperature coefficient of tungsten resistivity is  $\alpha = 4.5 \times 10^{-3}(\text{deg C})^{-1}$ . What is the resistance of the tungsten filament in the previous exercise if its temperature is increased from  $20^\circ\text{C}$  to a typical operating temperature of  $2850^\circ\text{C}$ ?